

CP-IMoE: Collaborative Prompt-Guided Interactive Mixture-of-Experts for Incomplete Multimodal Learning

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Abstract

Multimodal learning has achieved remarkable success across various domains. However, in the real-world medical scenarios, the incomplete modality problem frequently arises due to various factors, which leads to substantial performance degradation. In this work, we propose Collaborative Prompt-guided Interactive Mixture-of-Experts (CP-IMoE), a two-stage framework that is robust to missing-modality conditions. Stage 1 performs knowledge decomposition to pre-train four experts that explicitly capture modality-specific, redundant, and synergistic information, respectively. This comprehensive knowledge decomposition enables the model to perceive subtle yet critical information that is often overlooked, facilitating more effective multimodal integration. Stage 2 introduces a Collaborative Prompt (CLP) module that combines a static missing-aware prior with a representation-aware dynamic prompt. Specifically, this module mitigates the representation collapse problem in classic Mixture-of-Experts pipeline, with a balanced expert utilization leading to stable knowledge integration. CP-IMoE maintains competitive performance across three multimodal medical benchmarks under a wide range of missing ratios. Qualitative result shows that our CP-IMoE aligns with human intuition, where it relies on synergistic information when available and gradually shifts to unimodal evidence as missingness increases. Code is available at ¹.

1. Introduction

Multimodal learning has achieved remarkable success across various domains by integrating complementary in-

formation from different modalities, such as cross-modal retrieval [31, 34] and multimodal medical image analysis [2, 10, 12, 16, 18]. However, most existing multimodal learning approaches are built upon an ideal assumption that all modalities are complete. In reality, due to the factors such as data privacy, equipment malfunction, or limited resources, the problem of incomplete modalities frequently arises. It is particularly severe in the medical domain where data and ethical issues are highly sensitive [1]. Recent studies [23, 36, 37] have also confirmed that models trained under the assumption of complete modalities often exhibit poor robustness when confronted with missing modalities, as they rely heavily on paired multimodal data. Therefore, developing a robust multimodal model capable of handling incomplete modalities is of great practical significance.

A straightforward solution to the missing-modality problem is modality-imputation, where imputation-based methods attempt to reconstruct missing features using generative models [4, 8, 29, 33]. Despite their success in traditional computer vision tasks, these methods encounter critical obstacles in medical imaging. Owing to lack of the medical knowledge background, they tend to introduce substantial noise and incur heavy computational overhead [20]. Meanwhile, another line of research [25, 30, 40, 41] leverages sophisticated network architectures to align different modalities into a shared latent space, thereby mitigating the sensitivity to missing modalities. However, such approaches often suffer from limited scalability and poor generalization, as their intricate and rigid structures restrict practical deployment across diverse datasets [17, 28].

Recently, Mixture-of-Experts (MoE)-based approaches [17, 39] have emerged as a promising direction to address the incomplete-modality problem due to their flexible architecture and superior performance. These approaches adopt a unified framework that builds modality-specific experts

¹<https://github.com/lijing-coder/CP-IMoE>

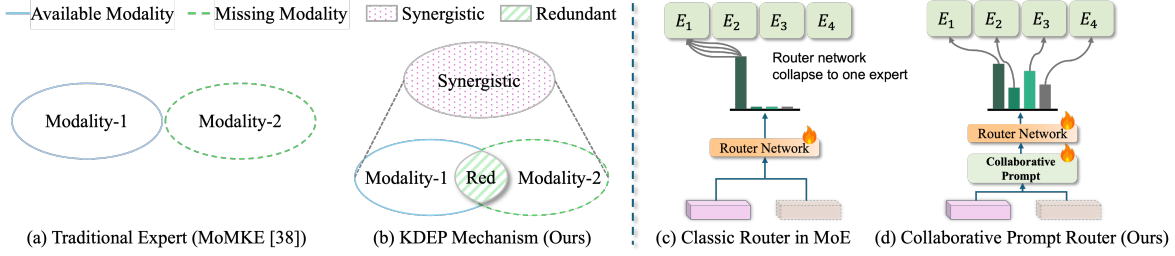


Figure 1. (a) Existing MoE-based multimodal learning methods mainly rely on unimodal experts and often neglect inter-modal interactions. (b) KDEP explicitly disentangles modality-specific, redundant, and synergistic components, respectively. (c) Classic router networks in MoE pipeline tend to suffer from representation collapse where routing weights degenerate to extreme values. (d) Our collaborative prompt-guided routing effectively reshapes the router’s input distribution, enabling balanced and stable expert knowledge utilization.

for discriminative feature learning, and rely on a router network to reassign weights to the available experts when a modality is missing. However, two major challenges remain. First, each modality-specific expert is constrained to capture unimodal information, while ignoring crucial inter-modal interactions, as shown in Fig. 1(a). For example, multimodal common knowledge may introduce redundancy, interfering with the learning of discriminative cues. Meanwhile, synergistic knowledge that the newly emerged information from multimodal interactions can easily be overlooked without explicit modeling. Second, the router network in MoE models often suffers from the representation collapse phenomenon [7, 24, 43]. Since the router input is derived directly from multimodal features with highly similar sample structures and concentrated distributions, and classic routers [14, 17, 39, 42] are often implemented as a single-layer MLP, they struggle to learn meaningful expert assignments. Consequently, the routing weights tend to degenerate toward extreme values (0 or 1), as illustrated in Fig. 1(c). This issue becomes even more pronounced under missing-modality scenarios [43].

To address the aforementioned challenges, we propose a Collaborative Prompt-guided Interactive Mixture-of-Experts (CP-IMoE) with a two-stage training framework. In the first stage, we introduce a knowledge decomposition expert pre-training (KDEP) mechanism (Fig. 1(b)) that systematically disentangles multimodal knowledge into four distinct experts, including the redundancy, synergy, and two modality-specific experts, respectively. Each expert is pre-trained to acquire specificity and interaction-aware feature representations. This comprehensive decomposition strategy facilitates a more effective integration of both modality-specific and inter-modality interaction information. In the second stage, a Collaborative Prompt-guided Interactive Mixture-of-Experts (CP-IMoE) network (Fig. 1(d)) is introduced to seamlessly integrate prompt learning with MoE. We regard different missing-modality cases as distinct expert-selection tasks and guide adaptive routing by introducing collaborative prompts. Specifically, we insert a

lightweight yet effective Collaborative Prompt (CLP) module between the input features and the router network. The CLP is composed of a static prompt that indicates which modalities are missing and a dynamic prompt that adaptively encodes representations from the available modalities. This design enables the router network to dynamically adjust its decision boundaries while compensating for information loss caused by missing modalities. Our model learns modality-specific and interaction-aware representations by leveraging the knowledge embedded in all pretrained experts. It demonstrates superior performance across three benchmark datasets over the state-of-the-art methods. The contributions are summarized as follows:

- We introduce a knowledge decomposition expert pre-training (KDEP) mechanism that systematically separates multimodal representation into modality-specific, redundant, and synergistic components, respectively. This leads to the expert explicitly modeling awareness of both unimodal cues and inter-modal interactions.
- We propose a Collaborative Prompt (CLP) module that integrates a missing-aware static prompt with representation-aware dynamic prompt through a collaborative strategy, enabling adaptive modality awareness under missing-modality conditions.
- CP-IMoE achieves state-of-the-art performance on three benchmark datasets while requiring only three trainable MLP layers during deployment. Qualitative analyses further demonstrate that CP-IMoE’s expert behavior aligns well with human intuition, highlighting the interpretability and robustness of the proposed design.

2. Related Work

Incomplete Multimodal Learning focuses on how to learn effectively when some modalities are missing. Imputation-based methods leverage generative models [4, 8, 29, 33] to reconstruct missing modalities from available ones. However, they often introduce noise and suffer from heavy computational overhead. Furthermore, researchers have

explored more complicated joint representation learning [25, 30, 40, 41], which captures the shared information across modalities to obtain cross-modal consistent representations. These methods can alleviate model sensitivity to missing modalities to some extent. However, their shared encoders are often structurally complex and rigid, which limits their practical applicability.

Mixture of Experts has shown remarkable success in various domains such as natural language processing [3, 19] and image processing [5, 6, 38], by leveraging the complementary strengths of multiple expert subnetworks. A typical MoE consists of multiple expert models and a gating network (router) that assigns weights to each expert. Recently, MoE-based frameworks have gained attention in addressing missing-modality problems due to their flexible architecture. For instance, MoMKE [39] pre-trains each unimodal expert via MLPs to learn modality-specific representations and performs downstream tasks using the remaining available experts when modalities are missing. However, its full-parameter training scheme results in high computational cost. Building upon this, SimMLM [17] proposes a More-vs-Fewer ranking loss that enhances model robustness by enforcing a ranking comparison between modality-rich and modality-poor pairs. Although these approaches achieve strong performance, they primarily focus on modality-specific representations, neglecting the inter-modal interactions that are essential for downstream tasks. Moreover, since their routers are implemented using only a single MLP layer and the samples exhibit highly similar structural patterns, they struggle to learn meaningful expert assignments. This issue becomes even more pronounced under missing-modality conditions [43].

In contrast, our approach simultaneously emphasizes modality specificity and inter-modality interactions through a comprehensive knowledge decomposition mechanism, which enhances the combination of multimodal knowledge. Additionally, by introducing a novel collaborative prompt module, our model allows the router to flexibly and robustly select appropriate experts, ensuring higher adaptability and stability—especially under missing-modality scenarios.

Prompt Learning aims to design or learn task-related prompts that guide models toward specific downstream objectives while tuning only a small number of parameters. This paradigm has been widely adopted in natural language processing [26, 27] and large-scale multimodal models [9, 15, 21, 35], demonstrating strong generalization across diverse tasks. Inspired by these advances, we incorporate prompt learning into the MoE framework for incomplete multimodal learning. We regard different missing-modality cases as distinct expert-selection tasks and introduce collaborative prompts to guide adaptive routing. Specifically, we combine static and dynamic prompts to enable the model to better handle missing modalities. By jointly leverag-

ing both types of prompts, our method ensures comprehensive awareness at both modality and sample level, achieving greater flexibility and adaptability.

3. Methods

3.1. Problem Definition

We provide a concise formulation of the missing-modality scenario in classification task in this work. For simplicity and generality, we focus on a setting with $M = 2$ modalities, denoted as m_1 and m_2 (e.g., image-1 and image-2), which can be naturally extended to more modalities. Given a multimodal dataset $\mathcal{D} = \{\mathcal{D}_c, \mathcal{D}_{m_1}, \mathcal{D}_{m_2}\}$, we define the *modality-complete* subset as: $\mathcal{D}_c = \{(x_{m_1}, x_{m_2}, y)\}$, where x_{m_1} and x_{m_2} denote the feature representations of modalities m_1 and m_2 , respectively, and y is the corresponding ground-truth label. The *modality-incomplete* subsets are defined as: $\mathcal{D}_{m_1} = \{(x_{m_1}, y)\}$, $\mathcal{D}_{m_2} = \{(x_{m_2}, y)\}$, which correspond to missing modality m_2 and missing modality m_1 , respectively. The goal is to design a robust model capable of handling missing-modality scenarios, ensuring accurate predictions even when only one modality is available.

3.2. Overall Architecture

The overall architecture of our Collaborative Prompt-guided Interactive Mixture-of-Experts (CP-IMoE) framework is illustrated in Fig. 2. The core idea is to enhance multimodal robustness under missing modality scenarios by decomposing multimodal knowledge and adaptively integrating these experts through prompt-guided routing.

In both training and testing, pretrained encoders are used to extract high-dimensional features from each modality, while the features of missing modalities are zero-filled to avoid interference with the model’s decision process. Specifically, training proceeds consist of two stages. In the first stage of knowledge decomposition expert pretraining (KDEP), we decompose the multimodal representation into four distinct expert pathways: redundancy, synergy, and specificity of the two modalities. The specific experts learn discriminative features unique to each modality; the redundant expert captures common cross-modal redundancies; the synergistic expert models the interactive knowledge from multimodal cooperation. In the second stage, collaborative prompt-guided expert integration, we introduce a lightweight yet effective prompt-guided routing mechanism to dynamically integrate pretrained experts. The router network is guided by two types of prompts: A static prompt that encodes missing-aware prior, enabling the network to recognize which modalities are missing; A dynamic prompt that adapts to the representation distributions of the available modalities, promoting more discriminative and robust integration. By integrating these prompts into

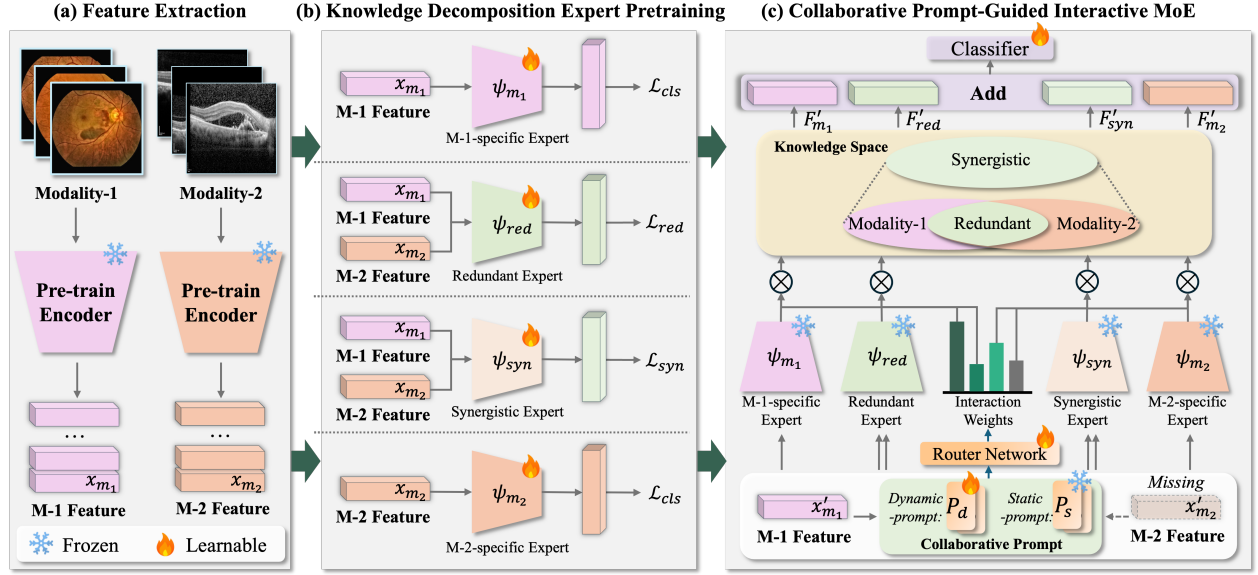


Figure 2. **Overview of CP-IMoE.** (a) The multimodal image features are extracted using pretrained encoders whose parameters remain frozen during training. (b) In Stage 1, we perform knowledge decomposition to pretrain four experts that capture modality-specific, redundant, and synergistic representations, respectively. (c) In Stage 2, collaborative prompts guide the router network to adaptively integrate experts under varying missing-modality conditions.



Figure 3. Expert activation on Harvard30k Glaucoma [22] with 40% of modality-2 missing. (a) Classic MoE router tends to collapse into assigning near-binary expert weights. (b) Our collaborative prompt-guided routing preserves adaptive expert integration.

the router network, the model avoids representation collapse and maintains adaptively adjust expert utilization even under incomplete modality conditions, as shown in Fig. 3.

3.3. Knowledge Decomposition Expert Pretraining

To achieve comprehensive knowledge decomposition, we employ a Mixture-of-Experts (MoE) architecture composed of four experts, including two interaction experts: ψ_{red} and ψ_{syn} , two specific experts: ψ_{m_1} and ψ_{m_2} . These experts are specifically designed to capture and model the distinct components that exist among multimodal features - namely, the *redundant*, *synergistic*, and *specific* components.

Specifically, we first adopt multilayer perceptron (MLP) layers as single-modality specific experts ψ_{m_1} and ψ_{m_2} to extract modality-specific feature F_{m_1} and F_{m_2} , which can

be formulated as:

$$F_{m_1} = \psi_{m_1}(x_{m_1}), F_{m_2} = \psi_{m_2}(x_{m_2}), \quad (1)$$

where x_{m_1} and x_{m_2} denote the extracted features from the two modalities. Both ψ_{m_1} and ψ_{m_2} are optimized for the pre-train classification task through a standard cross-entropy loss function $\mathcal{L}_{cls}$.

Unlike previous methods that did not distinguish between redundant and synergistic information [44], our framework explicitly distinguishes between these two types of multimodal relationships. We model multimodal interaction from a complementary perspective by leveraging a cross-modal co-attention mechanism to integrate knowledge across modalities. The interaction (redundant and synergistic) features C are formulated as:

$$C = \sigma(fc(\mathbf{A})) \cdot \phi_{m_1}(x_{m_1}) + \sigma(fc(\mathbf{A})) \cdot \phi_{m_2}(x_{m_2}), \quad (2)$$

$$\mathbf{A} = \phi_{m_1}(x_{m_1}) \odot \phi_{m_2}(x_{m_2}), \quad (3)$$

where σ denotes a sigmoid function, $fc(\cdot)$ is a fully connected layer, $\mathbf{A}$ represents the co-attention matrix, and $\phi(\cdot)$ denotes an MLP layer. The $\odot$ means matrix multiplication, while $\cdot$ indicates element-wise multiplication.

At the same time, to effectively disentangle redundancy and synergy, we introduce two distinct loss functions, e.g. $\mathcal{L}'_{red}$ minimizes the distance between multimodal features to capture common redundant information; $\mathcal{L}'_{syn}$ maximizes their discrepancy to emphasize synergistic representation, as shown in Fig 4. The two loss functions can be

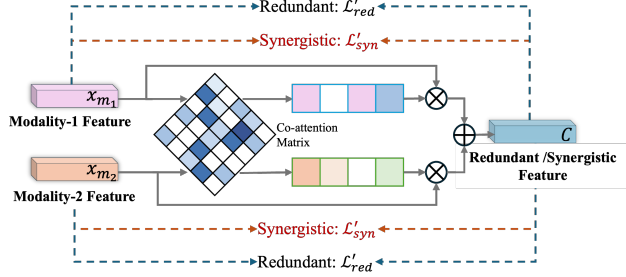


Figure 4. Architecture of the redundant and synergistic experts. The redundant and synergistic feature C is computed via attention-weighted integration based on the co-attention matrix and further enhanced by the redundant loss $\mathcal{L}'_{red}$ and synergistic loss $\mathcal{L}'_{syn}$, respectively.

formulated as:

$$\mathcal{L}'_{red} = \|x_{m_1} - x_{m_2}\|_2, \quad (4)$$

$$\mathcal{L}'_{syn} = \text{Max}(0, \delta - \|x_{m_1} - x_{m_2}\|_2), \quad (5)$$

where $\|\cdot\|_2$ represents the Euclidean distance, and δ is a margin parameter that controls the minimal separation between synergistic features. Thus, two interaction experts ψ_{red} and ψ_{syn} can be calculated by Eq. 2 and optimized based on $\mathcal{L}_{red}$ and $\mathcal{L}_{syn}$, respectively:

$$F_{red} = \psi_{red}(x_{m_1}, x_{m_2}), \quad F_{syn} = \psi_{syn}(x_{m_1}, x_{m_2}), \quad (6)$$

$$\mathcal{L}_{red} = \mathcal{L}'_{red} + \mathcal{L}_{cls}, \quad \mathcal{L}_{syn} = \mathcal{L}'_{syn} + \mathcal{L}_{cls}, \quad (7)$$

where F_{red}, F_{syn} are the output of two interaction experts. In summary, such comprehensive and explicit knowledge decomposition enhances the overall integrability and flexibility of the model in capturing multimodal relationship.

3.4. Collaborative Prompt-Guided Interactive Mixture-of-Experts

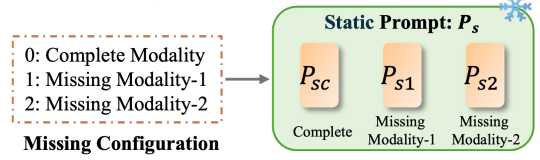
After the stage 1 pretraining, the experts acquire strong modality-specific and interaction-aware representation capabilities. At this stage, our objective is to integrate all these knowledge-enriched experts to better handle various incomplete multimodal scenarios.

3.4.1. Interactive Mixture-of-Experts Training

As illustrated in Fig. 2(c), the parameters of all pretrained experts are frozen to preserve their well-learned representational capacity and parameter sensitivity. The multimodal features are then fed into the four experts to obtain both modality-specific representations F'_{m_1}, F'_{m_2} and interaction representations F'_{red}, F'_{syn} , which can be formulated as:

$$F'_{m_1} = \psi_{m_1}^*(x'_{m_1}), \quad F'_{m_2} = \psi_{m_2}^*(x'_{m_2}), \quad (8)$$

(a) Static Prompt Generation:



(b) Dynamic Prompt Generation:

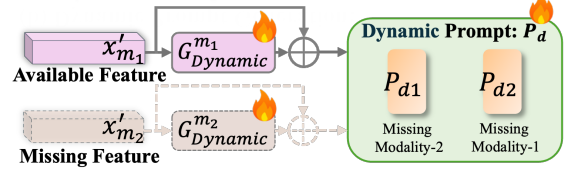


Figure 5. (a) Static prompt encodes missing-modality configuration as a structural prior, (b) Dynamic prompt adapts to feature information of available feature. Their combination provides both missing- and representation- awareness for stable routing.

$$F'_{red} = \psi_{red}^*(x'_{m_1}, x'_{m_2}), \quad F'_{syn} = \psi_{syn}^*(x'_{m_1}, x'_{m_2}), \quad (9)$$

where $*$ means the parameter of model is frozen. x'_{m_1}, x'_{m_2} denote the multimodal features used in the collaborative prompt-guided interactive mixture-of-experts training.

Due to the inherent complexity of incomplete multimodal data, the modality-specific and interaction representations may contain substantial semantic discrepancies. Fixed fusion strategies (e.g., simple weighted averaging) often fail to achieve robust performance under such conditions [39]. To overcome this limitation, we generate a static prompt P_s and a dynamic prompt P_d (Details can be found in Sec. 3.4.2) based on the available modalities and missing-modality configuration, such as $P_s, P_d = \text{CLP}(x'_{m_1}, x'_{m_2})$, where $\text{CLP}(\cdot)$ is the collaborative prompt generation module. These two prompts are passed through a router network, which dynamically estimates the importance weights of modality-specific and interactive representations. The final fused feature F'_{fus} is then obtained by integrating these weighted representations:

$$F'_{fus} = \sum_{k \in \{m_1, m_2, red, syn\}} w_k \cdot F'_k, \quad (10)$$

$$w_k = \text{Softmax}(W_1 \cdot \text{ReLU}(W_2 \cdot \text{Concat}(P_s, P_d))), \quad (11)$$

where w_k represents the weights of the four experts, W_1, W_2 are learnable parameters. Finally, the integrated representation is passed through a MLP layer for downstream classification. The CP-IMoE is optimized with a standard cross-entropy loss function $\mathcal{L}_{cls}$.

3.4.2. Collaborative Prompt Generation

Static Prompt for Missing Awareness: To handle incomplete multimodal inputs, we introduce a *static prompt* P_s

that explicitly encodes the missing modality configuration. The static prompt acts as a missing-aware prior, guiding the router to adapt its expert selection according to the presence or absence of each modality, as shown in Fig. 5(a). Specifically, given a modality availability vector $\mathbf{v} \in \{0, 1, 2\}$ indicating which modalities are available, P_s is formulated as an embedding conditioned on $\mathbf{v}$, such as $P_s = G_{Static}(\mathbf{v})$, where $G_{Static}(\cdot)$ is used to identify which modality is missing and encodes static prompt into a feature embedding represented by $\mathbf{v}$. During training, P_s is injected into the router to provide explicit awareness of the current modality configuration. This allows the model to avoid assigning excessive importance to the expert of missing modality. Additionally, P_s remains invariant at the same missing-modality condition, so it can be served as a structural prior to guide experts selection under incomplete modality conditions.

Dynamic Prompt for Representation Awareness:

While the static prompt encodes modality missing condition, it cannot capture the fine-grained semantic variations within the modalities. To address this, we propose a *dynamic prompt* P_d that adaptively percept the representation distribution of the available modality, as shown in Fig. 5(b). Given the available features x_{m_1} from the pretrained encoders, we compute the dynamic prompt as, $P_d = G_{Dynamic}^{m_1}(x_{m_1})$, where $G_{Dynamic}^{m_1}(\cdot)$ is a residual MLP module that dynamically captures the information of available modalities. Injecting P_d into the routing mechanism encourages the network to dynamically adjust expert integration weights based on the content of available modalities, leading to more robust experts selection. By combining P_s and P_d , the router network achieves both missing awareness and representation awareness, preventing expert dominance and promoting balanced knowledge integration under missing-modality conditions.

4. Experiments

4.1. Experimental Details

Datasets. We evaluate CP-IMoE on three multimodal medical benchmarks. **Harvard30k Glaucoma** [22] consists of paired *fundus* and *OCT* images for glaucoma screening. We use the binary classification setup (*normal* vs. *glaucoma*) with 1487 training pairs and 372 test pairs. **Derm7pt** [13] contains paired *clinical* and *dermoscopic* skin images. Following the standard Seven-Point Checklist protocol, we perform 5-class lesion classification, including *BCC*, *NEV*, *MEL*, *MISC*, and *SK*. The dataset includes 413 training pairs and 395 test pairs. **MMC-AMD** [32] provides paired *fundus* and *OCT* images for age-related macular degeneration diagnosis. We adopt the common 4-class setting: *wet AMD*, *dry AMD*, *PCV*, and *normal*, with 615 training pairs and 153 test pairs. Details are shown in the Supplementary Material.

Backbone. In our implementation, we extract image fea-

tures using ResNet50 [11], which is fine-tuned on the corresponding dataset. During deployment, both the image encoders and expert modules are frozen, and only the prompts, router, and classifier are tuned for the downstream task. In addition, to investigate the impact of backbone selection, we further evaluate a retinal foundation model (RETFound) [45] as an alternative backbone. The experimental results are reported in the Supplementary Material.

Implementation Details. All experiments were conducted on NVIDIA A100 GPU. The input images were standardized to a resolution of $3 \times 224 \times 224$. The batch size was set to 32, and the model was trained for 250 epochs using the Adam optimizer with a learning rate of $5e^{-5}$. During both training and testing, we simulate incomplete-modality scenarios according to a predefined missing rate p_{miss} . Specifically, for each sample, one or both modalities may be randomly dropped to mimic real-world scenarios. Three missing settings are considered: (1) *Modality-1 missing*, (2) *Modality-2 missing*, and (3) *random missing*, where the two modalities are alternately absent according to p_{miss} .

Evaluation Metrics. To evaluate the model’s performance, we employed three widely used classification metrics: F1-score, Cohen’s Kappa and Accuracy.

4.2. Comparison with State-of-the-art Methods

We compare the proposed CP-IMoE with five competitive incomplete multimodal learning baselines: **Modality Dropout (MD)**, which drops missing modalities during training process; **ShaSpec** [30], which learns shared feature representations across modalities; **MAPs** [15], which inserts modality-specific prompts into both the input and intermediate layers of the backbone; **MoMKE** [39], which adopts a Mixture-of-Experts framework to integrate modality-specific features; **RIMA** [41], which employs optimal transport to align multimodal features. Tab. 1 presents the comparison between our CP-IMoE and state-of-the-art multimodal learning methods under different missing-modality ratios (20%–80%) across the Harvard30k Glaucoma [22], Derm7pt [13], and MMC-AMD [32] datasets.

First, CP-IMoE achieves the best overall performance across all datasets and missing-modality ratios. On Harvard30k Glaucoma, it improves over the strongest baselines MAPs [15] and MoMKE [39] by 1.1%–2.3% in average F1 score, showing its ability to effectively capture both modality-specific cues and cross-modal synergy. Second, on the Derm7pt dataset, where lesion categories are diverse and fine-grained, CP-IMoE achieves a clear improvement, exceeding MAPs [15] by 3.74% F1 score under the 20% missing-modality setting, indicating strong robustness even on complex datasets. Third, on MMC-AMD dataset, CP-IMoE maintains stable performance even when over 60% of modalities are missing, while baseline methods MAPs [15] exhibit noticeable performance drops, demonstrating

Dataset	Methods	Missing Rate: 20%			Missing Rate: 40%			Missing Rate: 60%			Missing Rate: 80%		
		F1	Kappa	Acc	F1	Kappa	Acc	F1	Kappa	Acc	F1	Kappa	Acc
Harvard30k Glaucoma	MD	70.20	40.84	72.31	70.60	41.75	72.85	67.34	36.30	70.97	66.05	33.34	69.35
	ShaSpec [30]	70.35	41.23	72.58	69.28	39.97	72.58	70.51	41.06	71.77	68.41	37.15	70.43
	MAPs [15]	70.30	41.38	72.85	71.78	43.89	73.66	71.28	42.72	72.85	70.54	41.20	72.04
	MoMKE [39]	71.60	43.34	73.12	71.31	43.74	73.92	69.63	40.35	72.58	70.01	40.60	72.31
	RIMA [41]	70.88	42.07	71.24	70.54	41.20	72.04	69.21	38.44	70.43	67.92	35.84	69.09
	CP-IMoE (Ours)	72.74	45.61	74.19	72.82	45.66	73.92	72.31	44.70	73.66	72.94	45.94	74.19
Derm7pt	MD	56.27	52.41	72.66	56.06	52.39	72.91	58.08	51.90	72.66	56.18	49.79	71.14
	ShaSpec [30]	56.16	54.54	73.42	49.38	47.83	70.63	53.38	51.00	71.65	52.82	51.50	72.15
	MAPs [15]	59.44	56.56	74.94	60.82	58.68	75.70	60.55	57.06	74.68	60.21	56.18	74.18
	MoMKE [39]	53.15	54.54	73.67	54.15	48.22	69.87	53.16	51.03	71.14	48.24	46.29	68.86
	RIMA [41]	55.89	52.43	72.41	56.01	49.97	69.87	52.52	49.31	70.13	49.23	47.44	69.62
	CP-IMoE (Ours)	63.18	60.42	76.71	61.63	58.92	75.70	61.39	57.98	75.19	63.39	57.81	74.68
MMC-AMD	MD	84.73	77.70	84.97	81.67	72.79	81.70	81.56	70.54	80.39	78.79	67.83	78.43
	ShaSpec [30]	83.91	76.63	84.31	82.05	74.23	82.35	81.16	71.50	80.39	79.79	70.39	80.39
	MAPs [15]	84.37	77.57	84.97	81.22	74.90	83.01	79.81	72.81	81.70	80.31	73.05	81.70
	MoMKE [39]	85.49	78.60	85.62	80.60	75.44	83.01	78.06	72.43	81.05	80.40	73.50	81.70
	RIMA [41]	83.41	76.13	83.66	81.17	73.29	81.70	80.39	72.00	81.05	77.33	69.42	79.08
	CP-IMoE (Ours)	86.15	78.62	85.62	83.96	78.79	85.62	83.90	74.63	83.01	80.79	72.83	81.70

Table 1. Performance comparison under different missing-modality ratios on the Harvard30k Glaucoma [22], Derm7pt [13] and MMC-AMD [32] datasets. These experiments are conducted under random missing patterns. Bold number indicates the best performance.

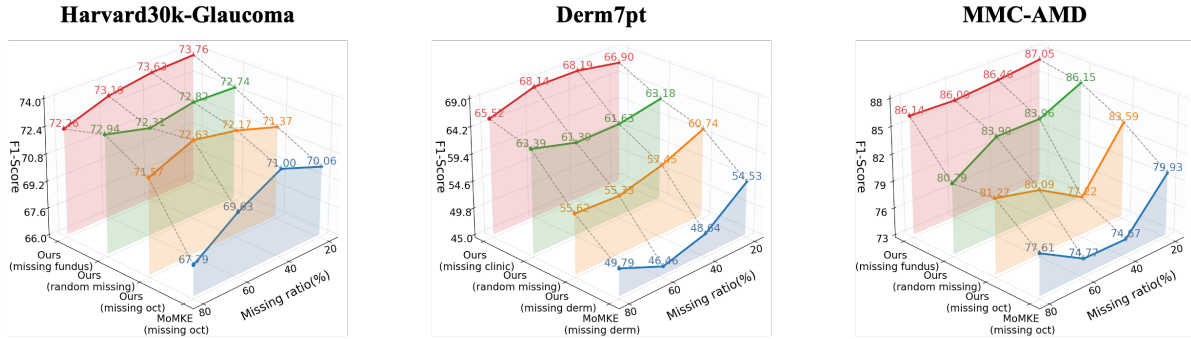


Figure 6. Robustness of CP-IMoE to different missing patterns on three datasets. Performance comparison under different missing-modality ratios, including single-modality missing and random missing. CP-IMoE maintains stable and superior performance under various missing-modality conditions, while the impact of missing different modalities varies across missing patterns.

superior resilience to severe modality incompleteness.

Overall, across experimental results of Tab. 1, CP-IMoE consistently improves F1/Kappa and maintains competitive Accuracy, with especially pronounced gains under high missing rates. For instance, at the 80% missing ratio, it outperforms the best baseline MAPs [15] by 2.4% and 3.18% in F1-score on Harvard30k Glaucoma and Derm7pt, respectively. These results substantiate the robustness of our model to incomplete modalities.

4.3. Ablation Study

Robustness to different missing configurations. To analyze the robustness of CP-IMoE, we evaluate its F1 score under different missing-modality ratios and missing com-

binations (single-modality missing and random missing) across all three datasets, as shown in Fig. 6. CP-IMoE consistently achieves clear and stable performance gains over MoMKE [39] across all missing configurations.

Visualization analysis of experts weight. To further understand the roles of modality-specific and cross-modal interactions under incomplete conditions, we visualize the expert load during the mixture-of-experts training on the Harvard30k Glaucoma, i.e., the router weights assigned to each expert on the test set. As shown in Fig. 7, the expert load gradually stabilizes as training converges. The visualization reveals a clear trend: when the missing-modality ratio is low, the model tends to rely more on the synergistic expert, as the cross-modal synergy provides richer discrimina-

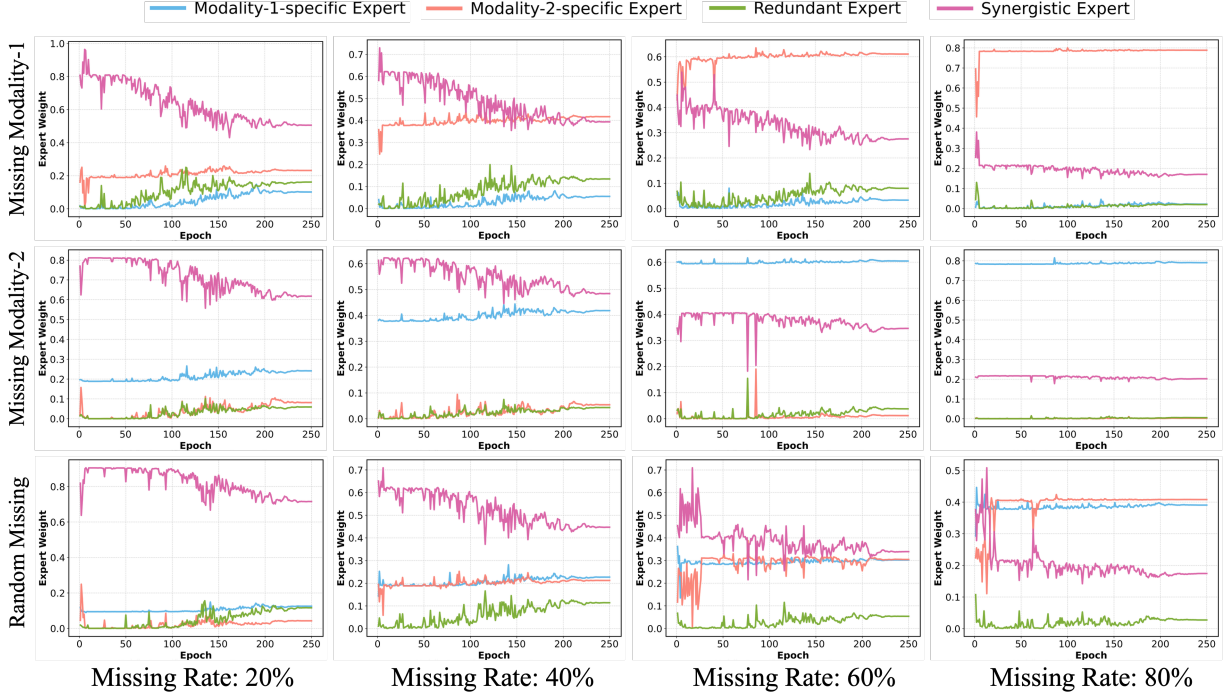


Figure 7. Expert weight visualization under different missing-modality ratios. Each subplot represents an incomplete case. The horizontal axis denotes the training epochs, the vertical axis is the expert weight after softmax.

Methods	Harvard30k (Kappa)		Derm7pt (F1 score)		MMC-AMD (Acc)	
	20%	80%	20%	80%	20%	80%
w/o specific expert	44.47	40.69	62.04	58.95	84.97	79.08
w/o interactive expert	44.08	41.44	61.67	59.99	84.31	79.74
w/o CLP	43.97	41.06	61.77	58.70	84.31	78.43
w/o static prompt	44.59	42.87	62.74	60.09	84.97	79.08
w/o dynamic prompt	44.98	42.21	63.02	59.45	85.62	78.43
CP-IMoE	45.61	45.94	63.18	63.39	85.62	81.70

Table 2. Ablation study of each components under missing-modality ratios of 20% and 80%.

tive cues. However, when the missing ratio becomes high (e.g., >60%), the synergistic information diminishes, and the model progressively shifts its reliance toward the available modality-specific experts (this is particularly evident in the first and second rows of Fig. 7). Moreover, the redundant expert is rarely selected across all settings, indicating that the model naturally avoids representations that do not contribute discriminative information. Encouragingly, this adaptive routing behavior aligns well with human intuition. When multimodal synergy weakens, reasoning reverts to the most reliable unimodal evidence.

Component Validation. As shown in Tab. 2, removing any key component leads to performance degradation. For example, when the modality-specific expert is removed,

the Kappa score on Harvard30k Glaucoma decreases from 45.61% to 44.47%, and removing the interactive expert lowers the classification accuracy on MMC-AMD from 85.62% to 84.31%. The collaborative prompt (CLP) also plays a critical role. On Derm7pt, removing the static prompt leads to a moderate decrease in F1 score from 63.18% to 62.74%, while removing CLP results in a larger decline to 61.77%, showing that our routing mechanism guidance is important.

5. Conclusion

In this paper, we introduced CP-IMoE, a collaborative prompt-guided interactive mixture-of-experts framework for incomplete multimodal learning. By explicitly decomposing modality-specific, redundant, and synergistic representations and guiding expert routing through static and dynamic prompts, our model effectively captures both modality-specific representations and inter-modal interactions, enabling adaptive and balanced utilization of all experts. Extensive experiments on three multimodal datasets show that our CP-IMoE outperforms state-of-the-art methods, demonstrating superior robustness and effectiveness across diverse missing ratios.

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